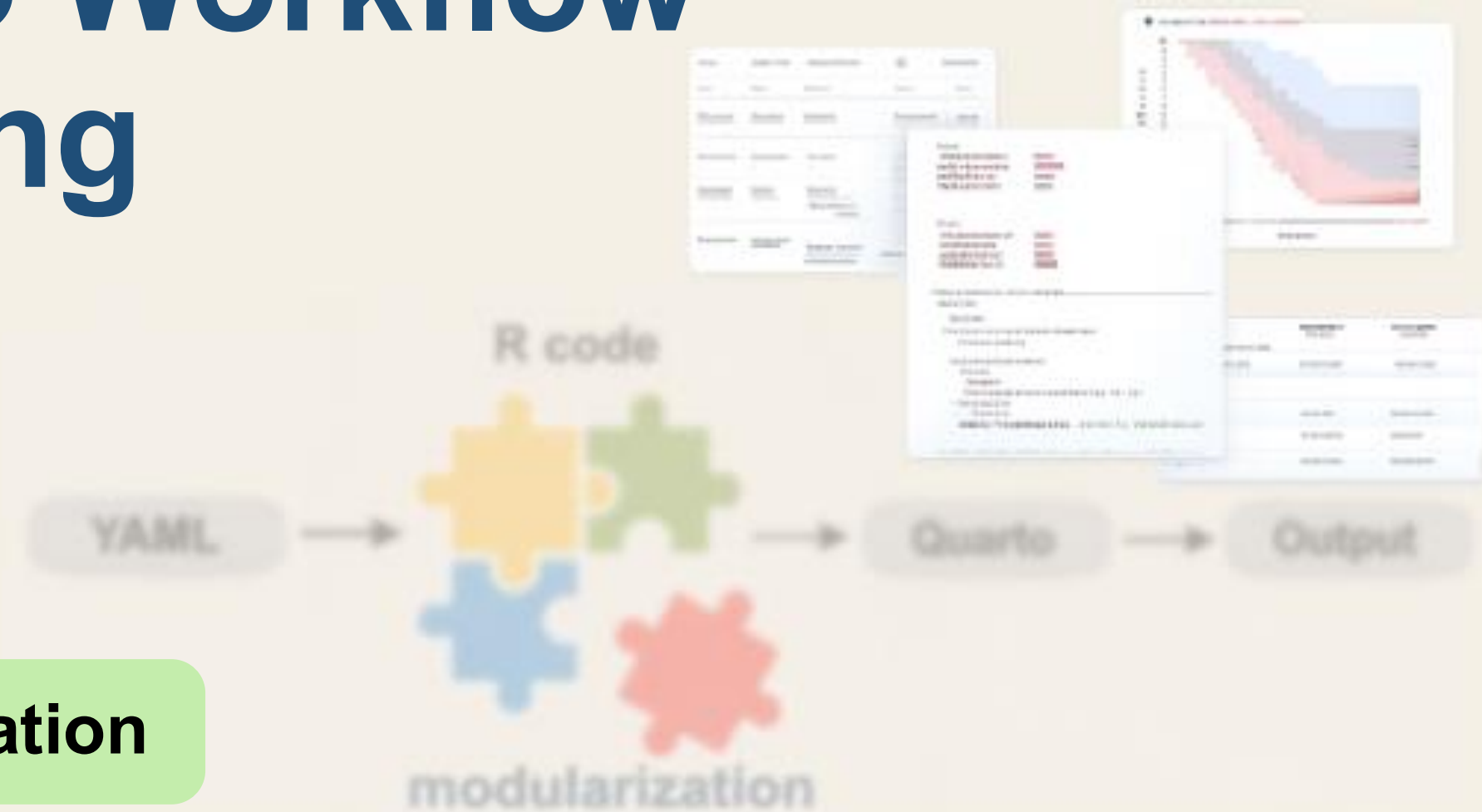


Reproducible Clinical Data Review: A Modular R and Quarto Workflow for Transparent Reporting

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Clinical Statistical Programming · Data Visualization

R + Quarto + YAML



**Reduced data
monitoring report
time by**

60%

with a modular workflow

Every new cut of the data
is a multi-day round trip

**Extract the
data**

**Map the
Variable**

**Write the
code**

Output

Send it back

***The back-and-forth
takes days***

***The output often
can't be traced back***

A modular workflow

Built in R + Quarto

```
1 # 1. Field Mapping
2 fields:
3     dm:
4         subject_col: PT
5         arm_col: ARM
6         site_col: SITE_ID
7         trt_start_col: TRTSDTC
8         pop_flag_col: SAFFL
9         age_col: AGE_FUL
10        sex_col: SEX_FUL
11        race_col: RACE_FUL
12        ethnic_col: ETHNIC_FUL
13        ecog_col: ECOG
14
15    ae:
16        subject_col: PT
17        term_col: AETERM_FUL
18        pt_col: AEPT_FUL
19        soc_col: AESOC_FUL
20        grade_col: AETOXGR_FUL
21        rel_col: AEREL_FUL
22        serious_col: AESER_FUL
23        ongoing_col: AEONGO_FUL
24        outcome_col: AEOUT_FUL
25        action_col: AEACN_FUL
26        start_col: AESTDAT_FUL
27        end_col: AEENDAT_FUL
28
```

```
# 2. Filters
```

```
filters:
```

```
  global:
```

```
    population: "SAFFL == 'Y'"
```

```
    sites: all
```

```
    subjects: all
```

```
  subject_criteria:
```

```
    combine: "OR"
```

```
    criteria:
```

```
      - label: "Has AE Grade >= 3"
```

```
        domain: ae
```

```
        condition: "as.numeric(GRADE_NUM) >= 3"
```

```
      - label: "Has abnormal ALT or AST"
```

```
        domain: lb
```

```
        condition: "LBTEST_FUL %in% c('ALT', 'AST') & LBNRIND_FUL == 'HIGH'"
```



```
98 # 4. Labels
99 listings:
100   - domain: ae
101     sort_by: [PT, AESTDAT_FUL]
102     columns:
103       - {col: PT,          label: "Subject ID"}
104       - {col: AETERM_FUL,  label: "AE Term"}
105       - {col: AEPT_FUL,    label: "Preferred Term"}
106       - {col: AESOC_FUL,   label: "SOC"}
107       - {col: AETOXGR_FUL, label: "Grade"}
108       - {col: AEREL_FUL,   label: "Relationship"}
109       - {col: AESER_FUL,   label: "Serious"}
110       - {col: AEACN_FUL,   label: "Action Taken"}
111       - {col: AEOUT_FUL,   label: "Outcome"}
112       - {col: AESTDAT_FUL, label: "Start Date"}
113       - {col: AEENDAT_FUL, label: "End Date"}
114   - domain: lb
115     flag_col: LBNRIND_FUL
116     sort_by: [PT, LBDAT_FUL, LBTEST_FUL]
117     columns:
118       - {col: PT,          label: "Subject ID"}
119       - {col: CPEVENT,     label: "Visit"}
120       - {col: LBCAT_FUL,   label: "Category"}
121       - {col: LBTEST_FUL,  label: "Lab Test"}
122       - {col: LBORRES_FUL, label: "Result"}
123       - {col: LBUNIT_FUL,  label: "Unit"}
124       - {col: LBNRIND_FUL, label: "Flag"}
125       - {col: LBCLSIG_FUL, label: "CS"}
126       - {col: LBDAT_FUL,   label: "Date"}
```

A self-contained report

A self-contained report

- Filter conditions
- Tables
- Figures
- Listings

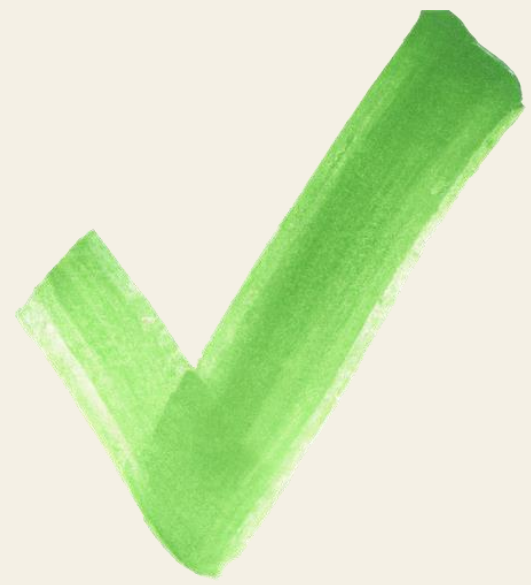
A new domain or chart ?

No touching the main code

Just adding a module

**time
saved**

60%+



Traceability



What did we
see at last check?

What conditions
were set?

NO
reconstruction

Clinical Data Review Report

PUBLISHED
June 25, 2026

Review Conditions

Table of contents

[Review Conditions](#)

- 1. Summary Statistics
- 2. Patient Timeline
- 3. CRF Listings

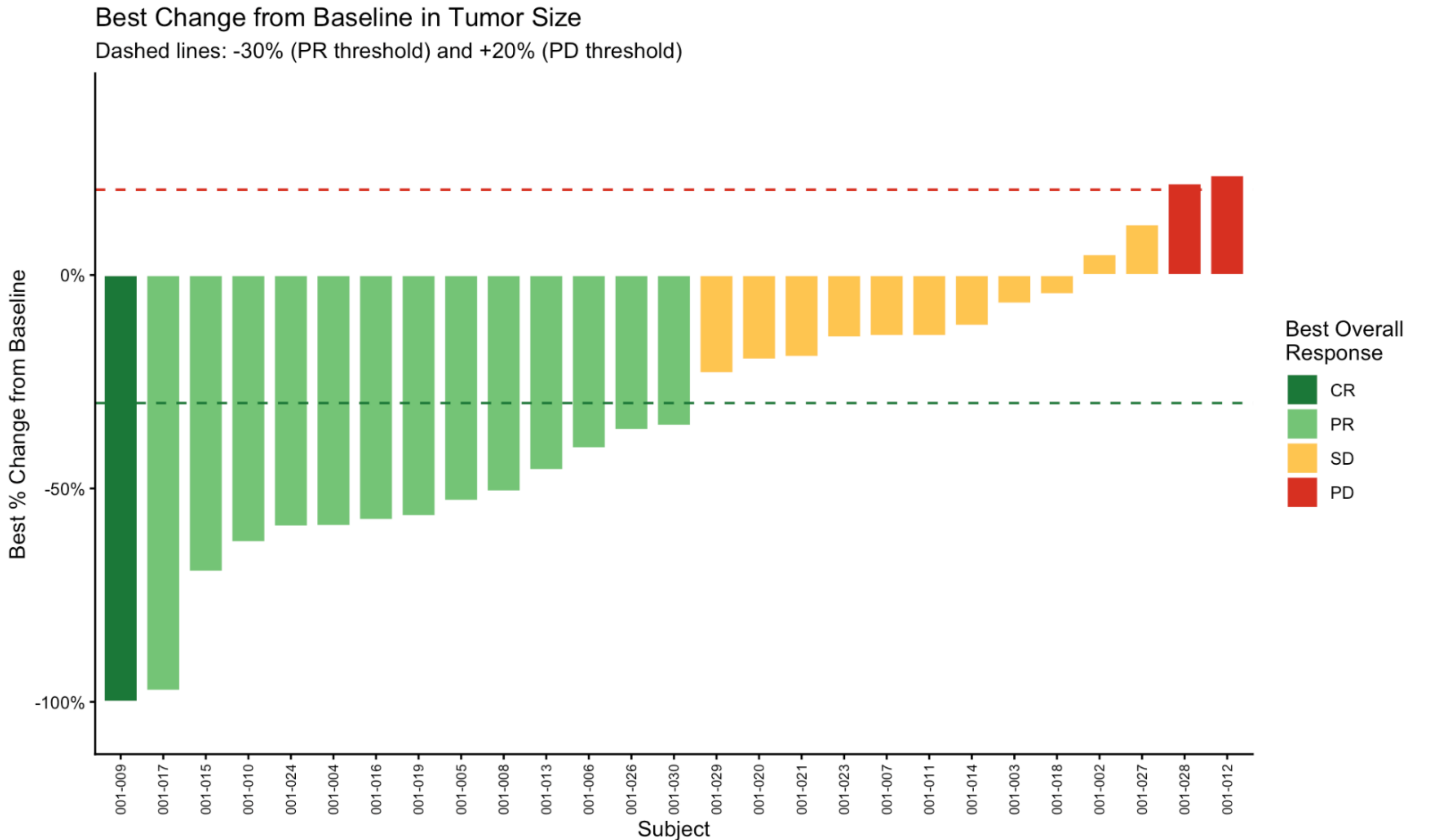
Active Review Conditions

Base Population: `SAFFL == 'Y'` – 30 subjects **Subject Criteria (OR):** - Has AE Grade $\geq 3 \rightarrow$ 16 subjects - Has abnormal ALT or AST $\rightarrow$ 27 subjects **Final Subject List:** 27 subjects *Generated: 2026-06-25 19:22:56 | Config: /Users/winkle/_Conference/A Modular R and Quarto Workflow for Transparent Reporting/config/config.yaml*

1.1 Demographic Characteristics

Characteristic	Treatment A N = 14 ¹	Treatment B N = 13 ¹	Overall N = 27 ¹
Age, years — Median (Min, Max)	60 (48, 69)	62 (53, 73)	61 (48, 73)
Sex			
F	2 (14%)	4 (31%)	6 (22%)
M	12 (86%)	9 (69%)	21 (78%)
Race			
ASIAN	4 (29%)	7 (54%)	11 (41%)
BLACK OR AFRICAN AMERICAN	2 (14%)	1 (7.7%)	3 (11%)
OTHER	1 (7.1%)	1 (7.7%)	2 (7.4%)
WHITE	7 (50%)	4 (31%)	11 (41%)
ECOG PS			
0	5 (36%)	5 (38%)	10 (37%)
1	7 (50%)	6 (46%)	13 (48%)

1.3 Best Change from Baseline in Tumor Size



3. CRF Listings

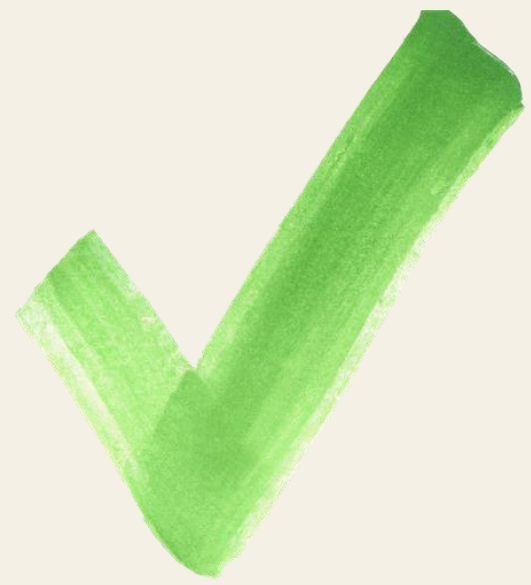
3.1 Adverse Event Listing

Copy

CSV

Search:

Subject ID	Treatment	Sex	AE Term	Preferred Term	SOC	Grade	Relationship
AI	All		All	All	All	All	All
001-002	Treatment A	M	Nausea	Nausea	Gastrointestinal disorders	Grade 1 - Mild	PROBABLY RELATED
001-002	Treatment A	M	Arthralgia	Arthralgia	Musculoskeletal and connective tissue disorders	Grade 1 - Mild	PROBABLY RELATED
001-003	Treatment A	M	Fatigue	Fatigue	General disorders and administration site conditions	Grade 2 - Moderate	PROBABLY RELATED
001-004	Treatment A	M	Arthralgia	Arthralgia	Musculoskeletal and connective tissue disorders	Grade 1 - Mild	PROBABLY RELATED

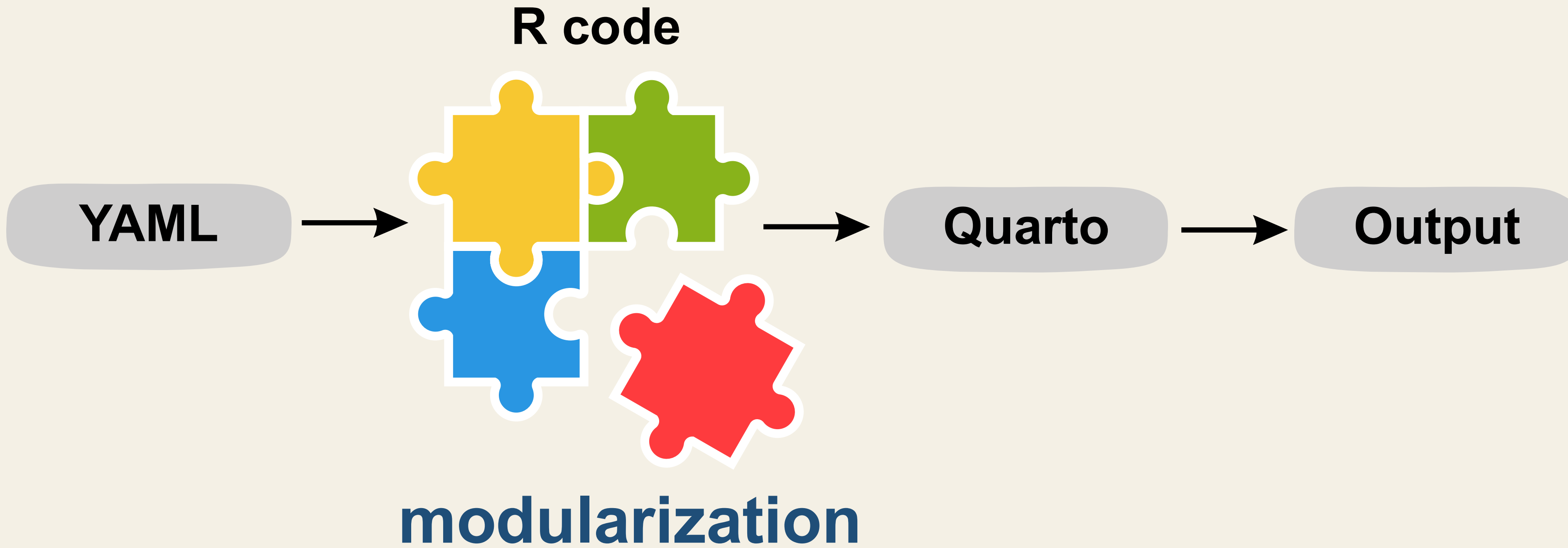


Reusable

Minimal code changes

Mostly YAML configuration

To Sum Up



Full article by Winkle

